

TRAVIS AFB, CA, USA

WMO: 745160

Lat: 38.267N

Lon: 121.933W

Elev: 19

StdP: 101.10

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23202

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.0	0.6	-8.2	1.9	12.1	-6.0	2.3	11.2	13.0	11.2	11.6	11.0	1.7	30	0.641

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	16.7	37.4	19.7	35.2	19.2	33.0	18.7	21.1	34.2	20.2	32.7	19.5	31.3	4.6	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.4	11.7	22.4	15.6	11.1	21.9	15.0	10.7	21.6	60.7	34.3	57.8	32.9	55.1	31.2	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.9	12.0	11.2	DB	-4.0	41.0	2.4	1.6	-5.8	42.1	-7.1	43.0	-8.5	43.9	-10.2	45.0	
			WB	-4.5	22.9	2.3	1.5	-6.2	24.0	-7.5	24.9	-8.8	25.7	-10.5	26.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.1	8.7	10.5	13.0	14.8	17.8	21.2	22.7	22.8	21.9	18.0	12.4	8.7
	DBStd	5.95	2.68	2.50	2.76	3.28	3.33	3.29	2.72	2.65	3.01	2.95	3.16	2.93
	HDD10.0	158	56	22	6	2	0	0	0	0	0	0	13	59
	HDD18.3	1379	297	219	168	114	49	8	1	1	4	42	180	298
	CDD10.0	2374	17	37	98	146	243	334	393	397	358	247	84	20
	CDD18.3	552	0	0	1	8	34	92	135	139	112	31	1	0
	CDH23.3	6746	0	1	29	170	517	1217	1516	1506	1304	454	32	1
(14) Wind	WSAvg	5.4	3.3	3.8	4.4	5.4	6.8	7.4	8.2	7.6	6.0	4.5	3.3	3.3
	PrecAvg	550	104	102	72	31	20	6	1	1	3	27	52	110
	PrecMax	899	339	328	255	107	79	50	10	14	14	128	160	304
	PrecMin	284	0	0	1	2	0	0	0	0	0	0	0	5
	PrecStd	156	90	74	69	26	20	12	2	3	4	30	38	85
	DB	20.0	22.3	26.3	31.0	35.2	39.0	39.3	39.3	37.9	33.9	26.8	19.9	
	MCWB	11.5	14.2	15.9	17.6	19.0	20.8	20.4	20.4	19.0	17.4	15.3	12.3	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	17.4	19.8	23.7	27.7	31.9	36.2	36.3	36.4	36.0	30.8	23.6	17.3	
	2% MCWB	12.2	13.6	15.3	16.6	18.2	20.1	19.8	19.8	18.6	16.8	14.7	12.1	
	5% DB	15.7	17.7	21.8	25.1	28.9	33.0	33.7	33.8	33.4	27.9	21.1	15.7	
	5% MCWB	11.4	12.2	14.5	15.6	17.0	18.8	19.5	19.2	17.9	15.6	13.5	11.4	
	10% DB	13.9	16.0	19.5	22.4	26.0	30.2	31.2	31.1	30.9	25.8	18.8	14.0	
	10% MCWB	10.7	11.6	13.5	14.5	16.0	18.0	18.8	18.8	17.4	14.9	12.8	10.9	
	WB	15.5	16.1	17.6	19.7	20.6	22.4	22.0	21.9	20.9	19.2	17.4	15.8	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	16.8	19.3	23.4	28.2	32.8	36.1	36.6	36.1	33.6	29.2	21.7	17.2	
	2% WB	13.5	14.9	16.3	17.8	19.0	20.7	20.7	20.7	19.7	18.0	16.2	14.2	
	2% MCDB	15.4	18.0	21.9	25.4	30.0	33.3	34.2	33.6	32.0	27.4	21.1	15.4	
	5% WB	12.4	13.7	15.2	16.4	17.8	19.6	19.8	19.8	18.9	16.9	15.1	13.0	
	5% MCDB	14.2	16.4	20.2	23.4	27.5	31.9	32.4	32.0	30.7	24.9	19.1	14.2	
	10% WB	11.4	12.5	14.1	15.2	16.6	18.4	19.0	18.9	18.0	16.1	14.0	11.8	
	10% MCDB	13.1	14.9	18.2	21.1	24.7	29.2	30.4	30.3	29.1	23.1	17.5	13.2	
(35) Mean Daily Temperature Range	MDBR	9.5	10.4	11.8	13.0	14.1	15.8	16.7	16.7	16.4	14.5	12.2	9.9	
	5% DB	13.2	13.8	15.7	17.7	18.9	20.5	20.3	20.8	20.9	18.8	16.3	12.2	
	5% MCWBR	8.4	8.3	8.8	8.7	7.6	7.5	7.0	7.1	7.0	7.8	8.4	7.7	
	5% MCDBR	9.2	11.3	13.6	15.5	17.4	19.0	18.8	18.6	17.9	15.6	12.9	7.7	
	5% MCWBR	6.9	7.6	8.1	8.3	7.7	7.5	7.1	7.0	6.6	7.0	7.6	5.9	
(40) Clear-Sky Solar Irradiance	taub	0.320	0.322	0.335	0.355	0.360	0.359	0.360	0.346	0.345	0.329	0.323	0.326	
	taud	2.526	2.524	2.473	2.398	2.408	2.417	2.407	2.437	2.445	2.520	2.518	2.511	
	Ebn at Noon	863	910	928	923	919	917	912	920	903	889	850	827	
	Edh at Noon	77	87	101	116	117	116	116	110	103	86	77	74	
(44) All-Sky Solar Radiation	RadAvg	2.25	3.26	4.63	6.11	7.32	8.12	7.99	7.14	5.86	4.18	2.75	2.00	
	RadStd	0.38	0.42	0.46	0.51	0.44	0.33	0.23	0.22	0.20	0.27	0.19	0.32	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	-0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (49 neighbors)	+0.38	N/A	-1.12	N/A	N/A	N/A	N/A	-117	+131	N/A

Nomenclature: See separate page